

Network Analysis:

The Hidden Structures behind the Webs We Weave

17-338 / 17-668

Homophily and Degree Correlation

Thursday, September 18, 2025

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The natural sciences perspective

Homophily: Status & Power

Degree homophily: “degree assortativity” or “degree correlation” – high-degree nodes tend to be connected to other high-degree nodes and vice versa.

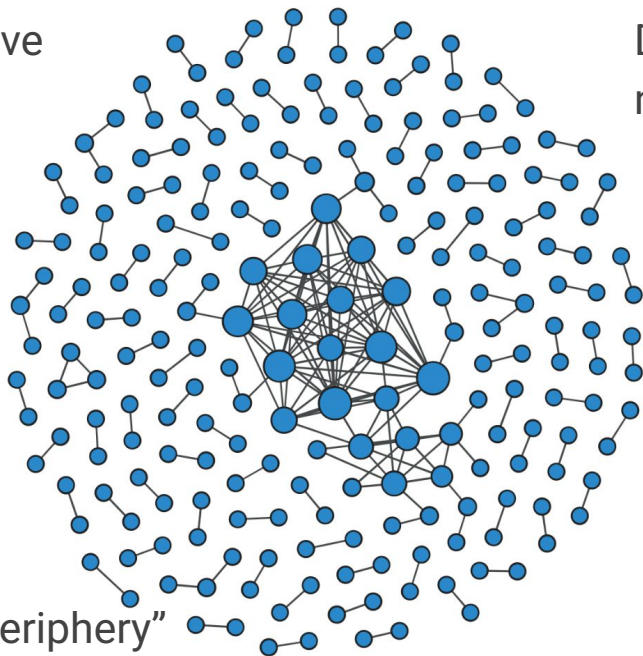
Extensively studied from a graph-theoretic perspective.



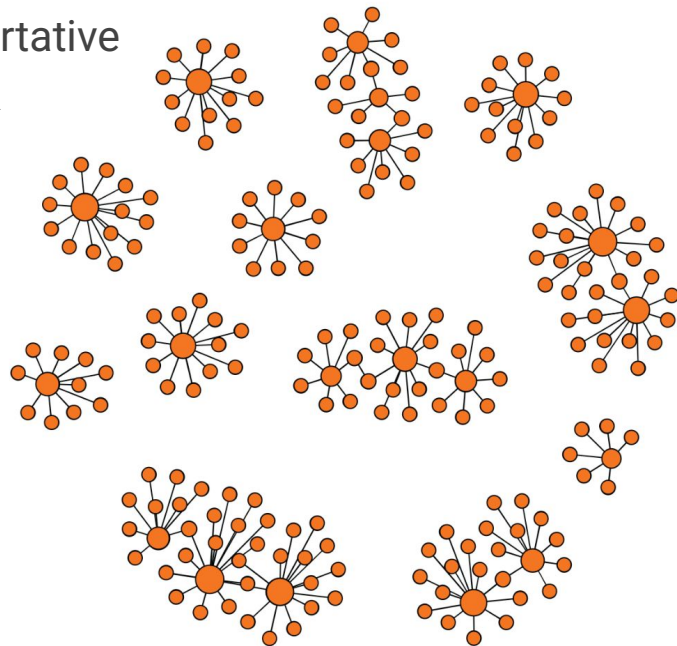
Degree Assortativity / Disassortativity

Example:

Assortative
network



Disassortative
network

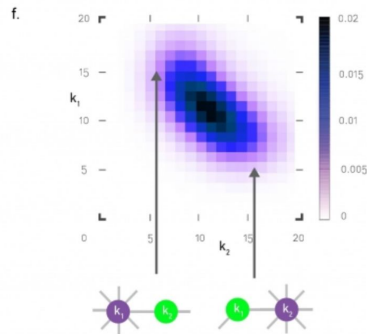
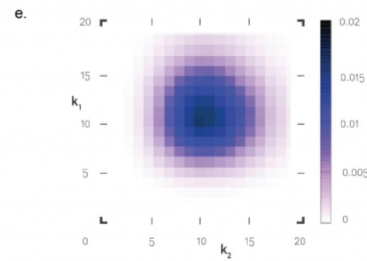
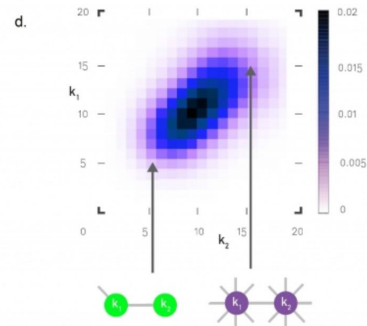
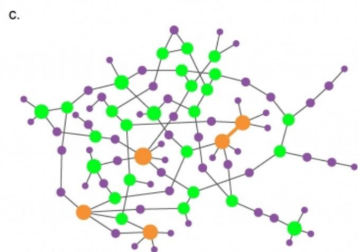
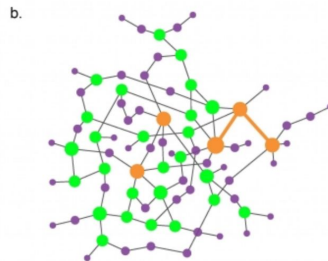
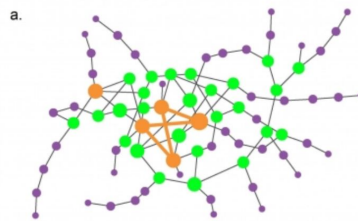


Degree Assortativity / Disassortativity

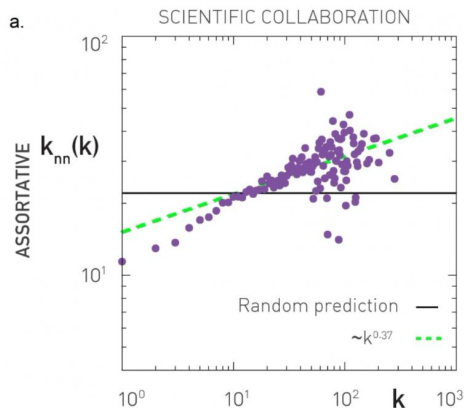
(a) **Positive** degree correlation: Connected nodes have similar degree

(a) **Neutral**: The degree of connected nodes have no correlation

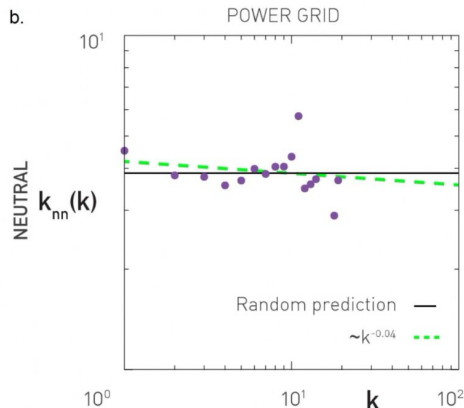
(a) **Negative** degree correlation: Connected nodes have dissimilar degree



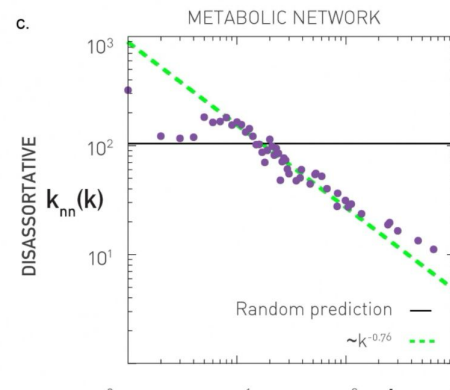
Measuring degree correlation: Average degree of the neighbors of a node of degree k



Average degree of neighbors increases as k increases → assortative network



Average degree of neighbors neither increases nor decreases as k increases → degree neutral network

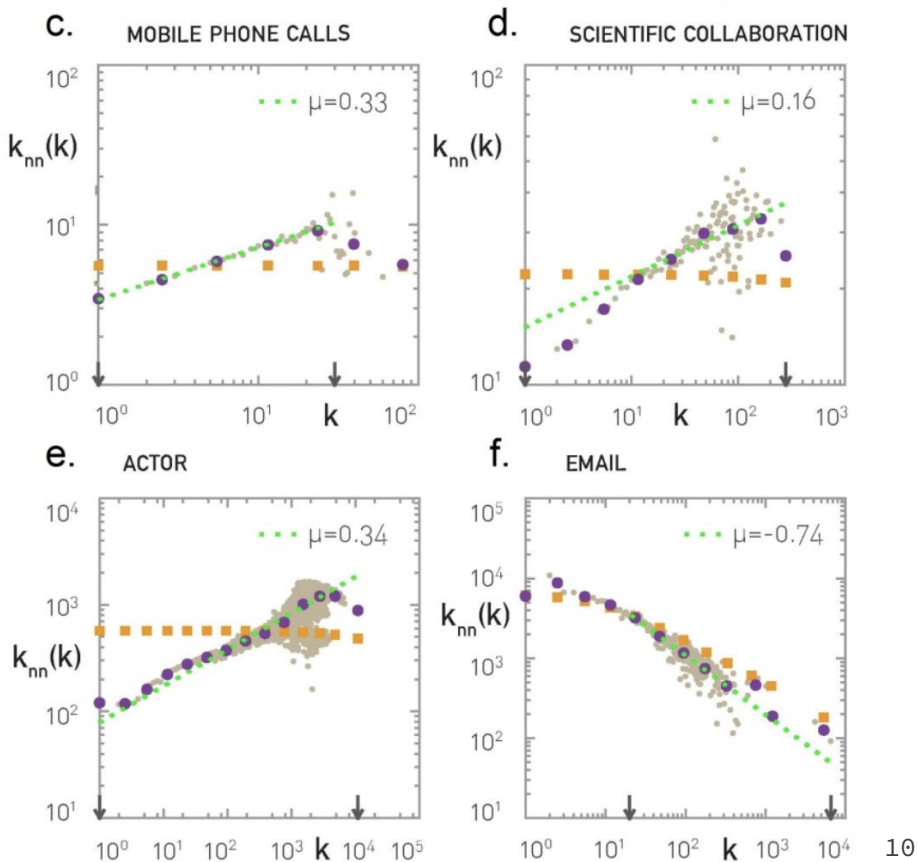


Average degree of neighbors decreases as k increases → disassortative network

Human social networks tend to exhibit positive degree correlations

Why positive?

Why is the email network negative?



Human social networks tend to exhibit positive degree correlations

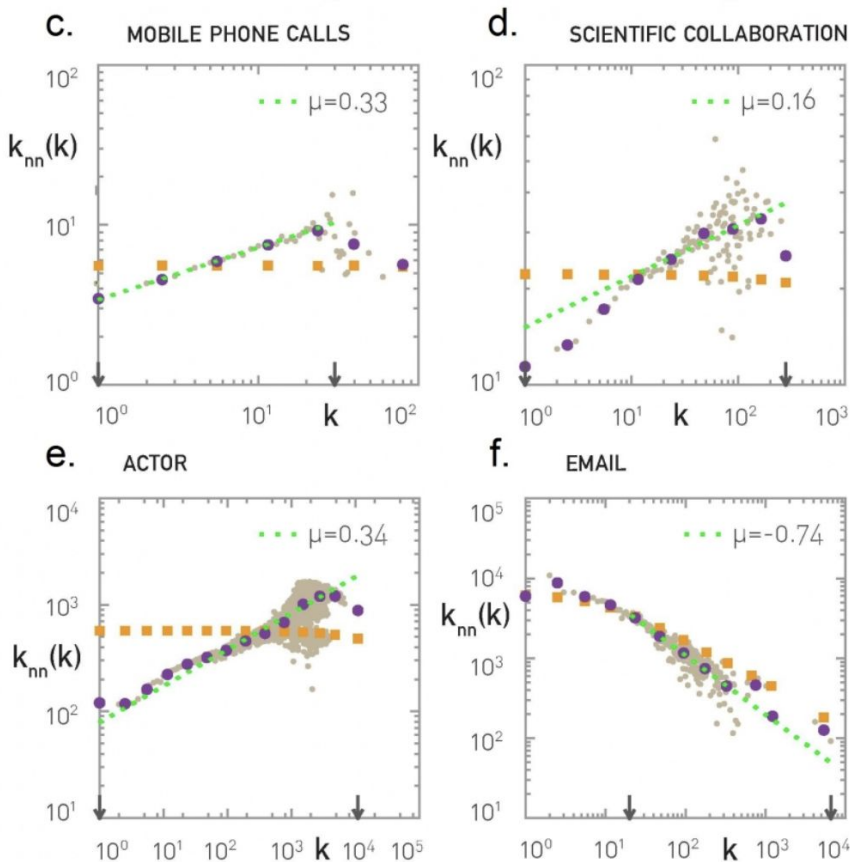
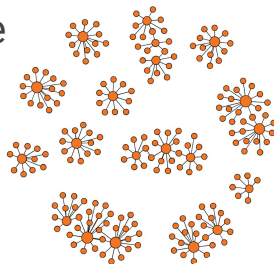
Why positive?

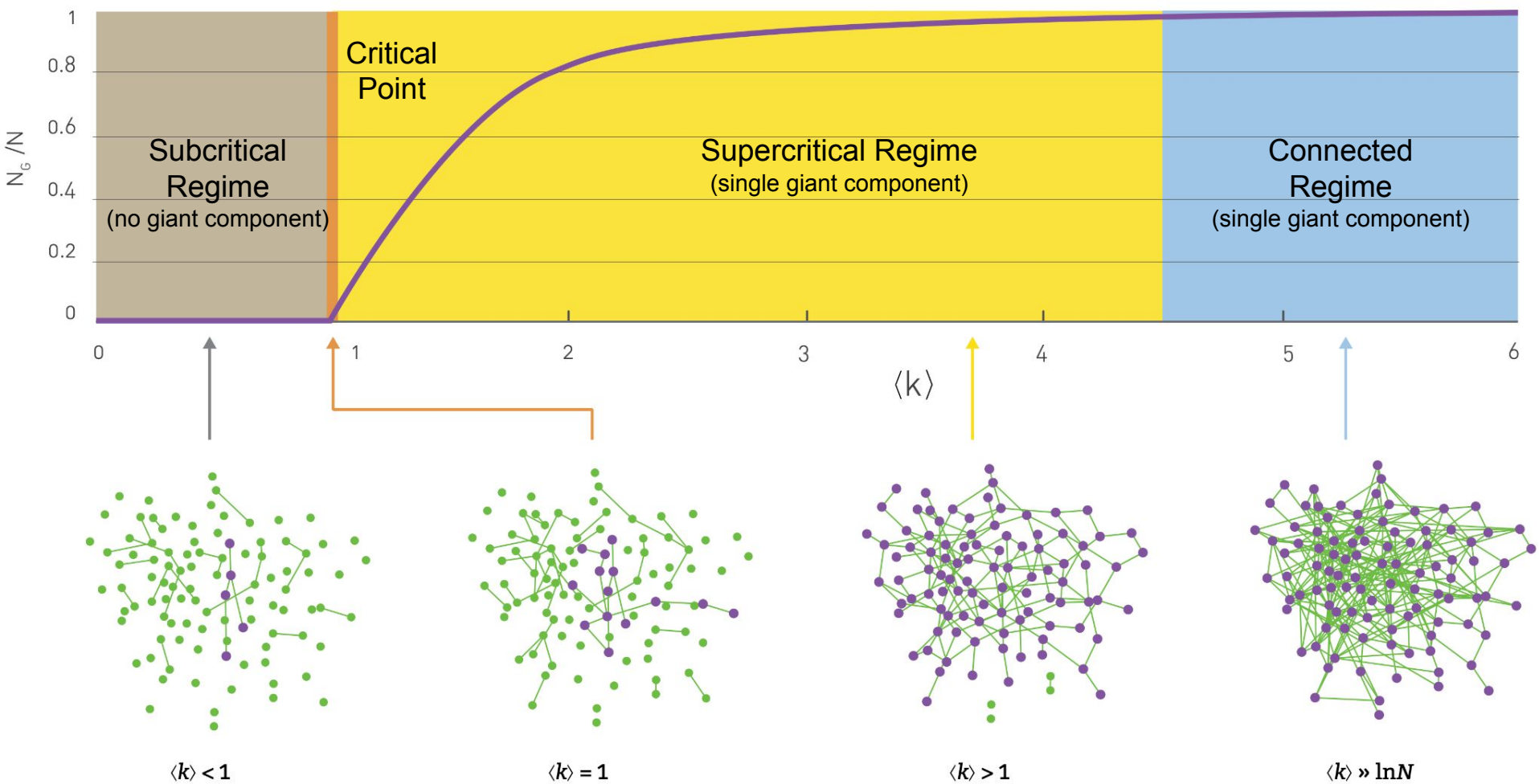
→ Open question. Several studies argue that it is related to the fact that humans form groups

→ People in large groups tend to have high degree (more group members to connect with) and those in small groups are constrained in forming ties - hence low degree

Why is the email network negative?

→ Networks with skewed degree distributions tend to exhibit negative degree correlations



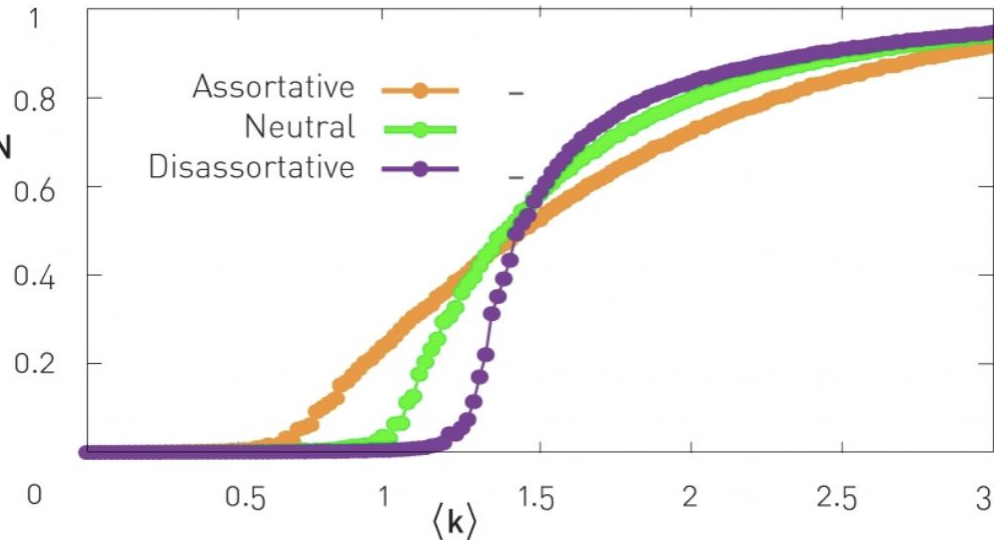


Impact of Assortativity: Higher connectivity

Giant component can emerge at lower mean degree $\langle k \rangle$

Size of largest
component /
Size of entire
network

→ S/N

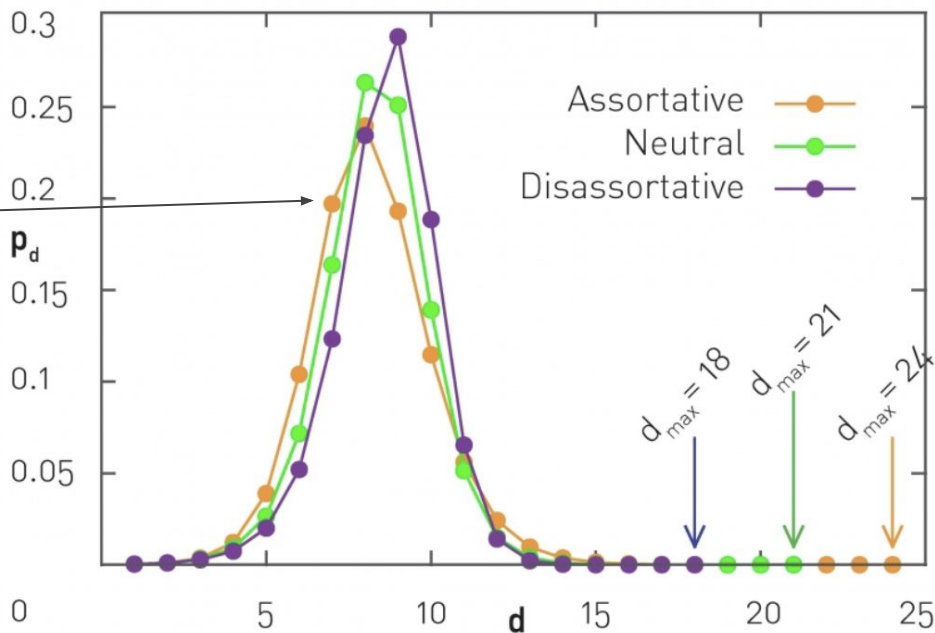


This means connectivity
increases even if people do not
have many connections

Impact of Assortativity: Higher connectivity

Giant component can emerge at lower mean degree $\langle k \rangle$

Assortative
networks have
shorter average
path length



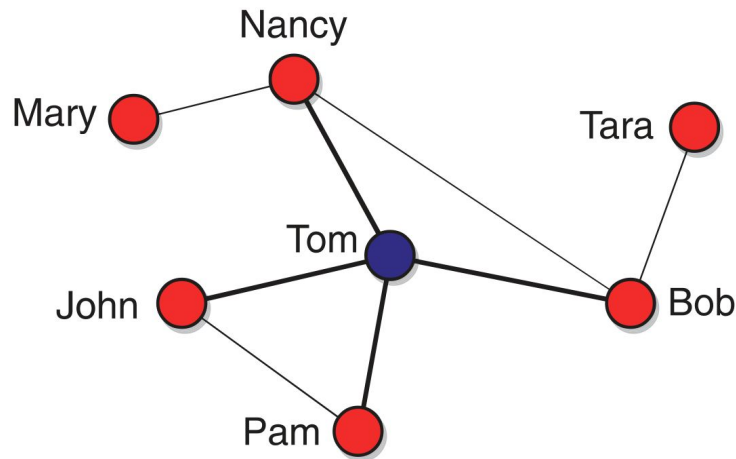
Case Study: The Friendship Paradox

Suppose you are looking for the person with the most friends

You only have a directory of phone numbers

Option 1: Call a person randomly

The chance that you pick Tom is ... ?

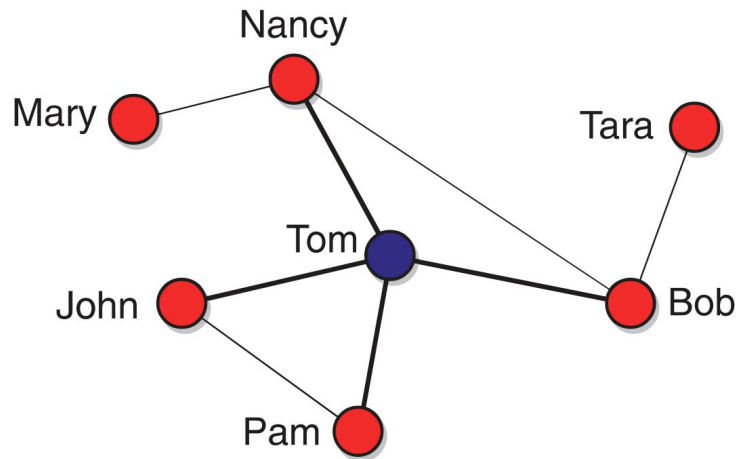


Suppose you are looking for the person with the most friends

You only have a directory of phone numbers

Option 1: Call a person randomly

The chance that you pick Tom is $1/7 \sim 14\%$

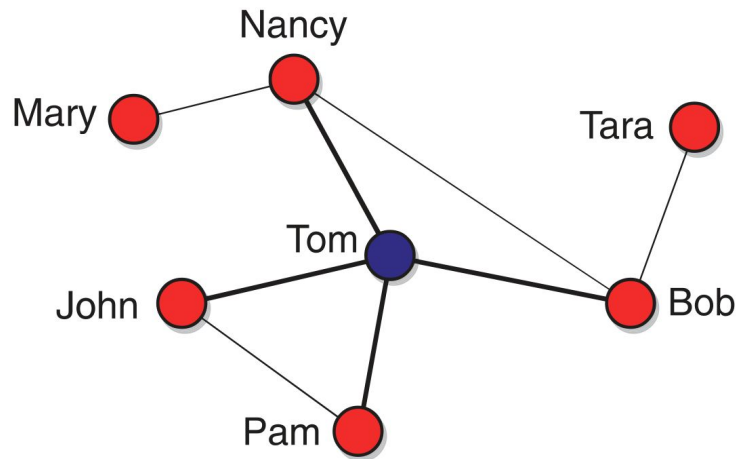


Suppose you are looking for the person with the most friends

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Option 2: Call a person randomly, and ask them about a random friend

The chance that you pick Tom is ...?



Suppose you are looking for the person with the most friends

You only have a directory of phone numbers

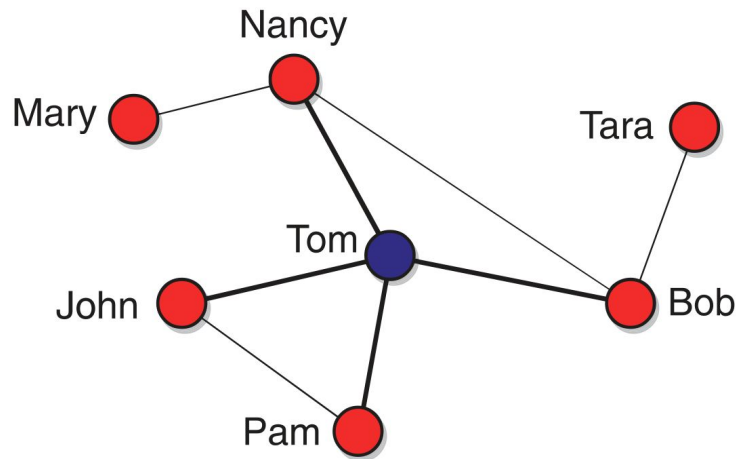
Option 2: Call a person randomly, and ask them about a random friend

The chance that you pick Tom is $5/21 \sim 24\%$

Mary: 0/1, Nancy: $\frac{1}{3}$, John: $\frac{1}{2}$, Pam: $\frac{1}{2}$, Bob: $\frac{1}{3}$, Tara: 0/1, Tom: 0/4

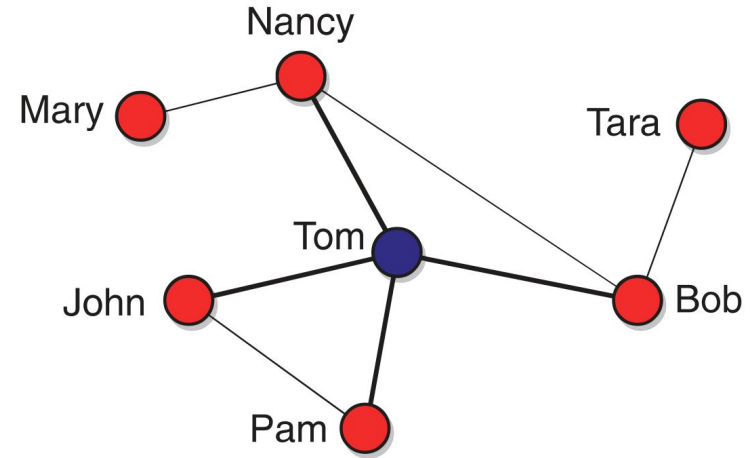
Probability of being called: $1/7$

Therefore: $(0/1 + \frac{1}{3} + \frac{1}{2} + \frac{1}{2} + \frac{1}{3} + 0/1 + 0/4) * 1/7 = 5/21$



Now, the paradox:

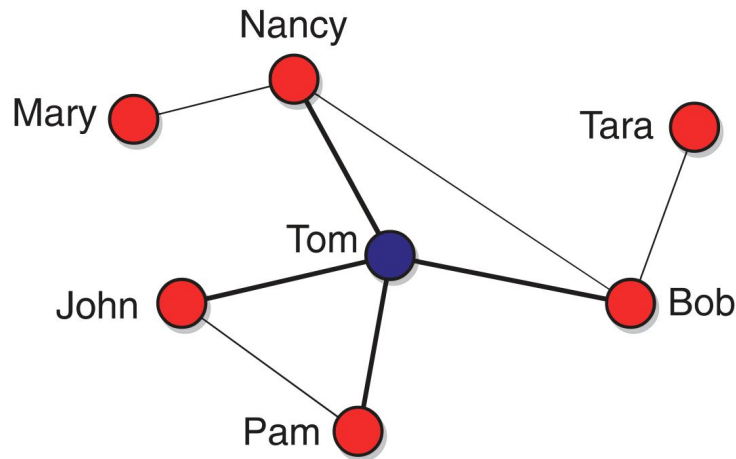
Average degree: ?



Now, the paradox:

Average degree: $(1+3+4+2+2+3+1)/7$
 $= 16 / 7 = 2.29$

Average degree of neighbors: ?



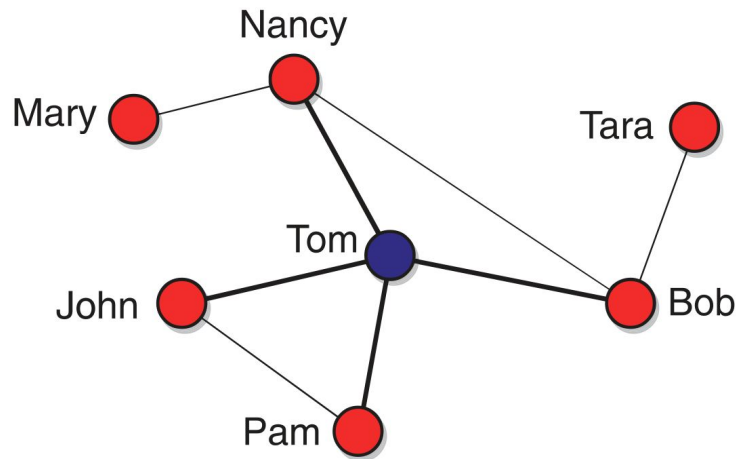
Now, the paradox:

Average degree: $(1+3+4+2+2+3+1)/7$
 $= 16 / 7 = 2.29$

Average degree of Toms Neighbors:

- Nancy: 3, John: 2, Pam: 2, Bob: 3
- $(3+2+2+3)/4=2.5$

Average of the average degree of neighbors:
 $(3+8/3+10/4+3+3+8/3+3) / 7 = 2.83$

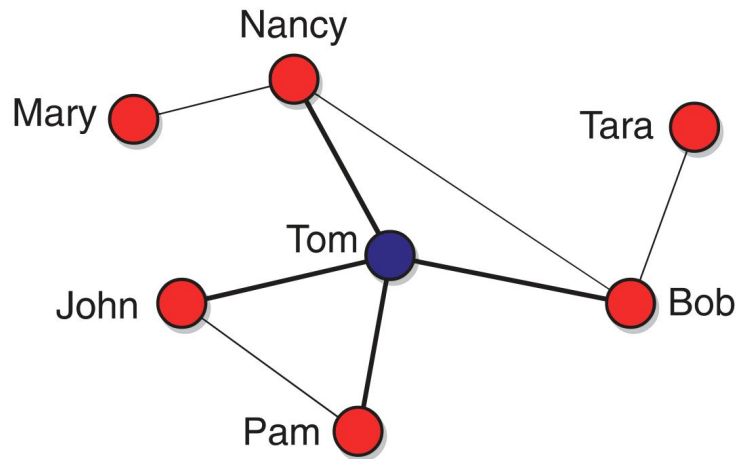


Now, the paradox:

Average degree: 2.29

Average degree of neighbors: 2.83

**Your friends have more friends than you,
on average!**

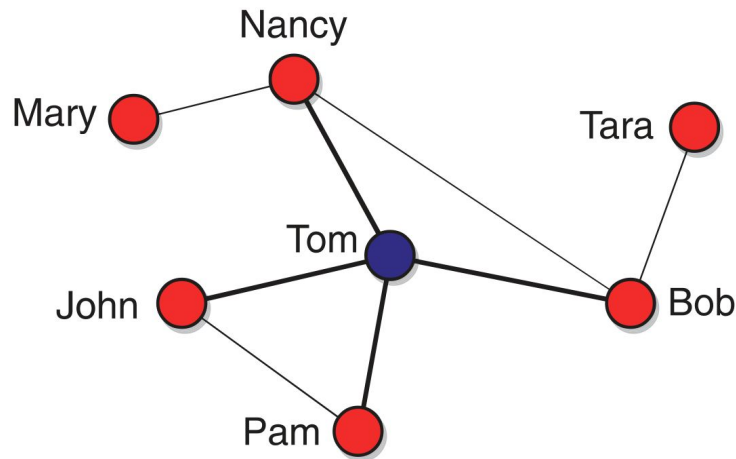


But it doesn't hold for everyone:

Nancy has 3 friends: Mary, Tom, Bob

They have in total $1 + 4 + 3 = 8$ friends

→ Nancy's friends have on average $8/3$ friends (i.e., less than Mary)



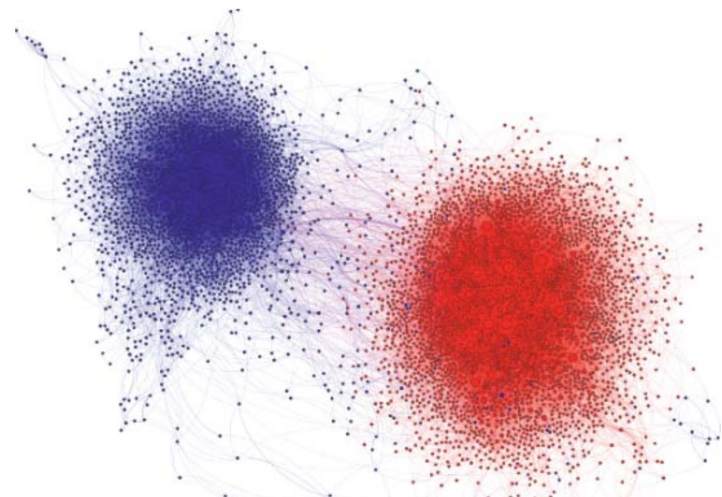
Aside: The dark side of homophily

Exceedingly easy to connect with people who share our worldviews and unfriend / unfollow people with different opinions.

Information can be shared and consumed in such a selective and efficient way as to influence our opinions very effectively.

Result: segregation and polarization of our online communities.

High risk of manipulation by misinformation and social bots.



Aside: Networks can also exhibit inverse homophily

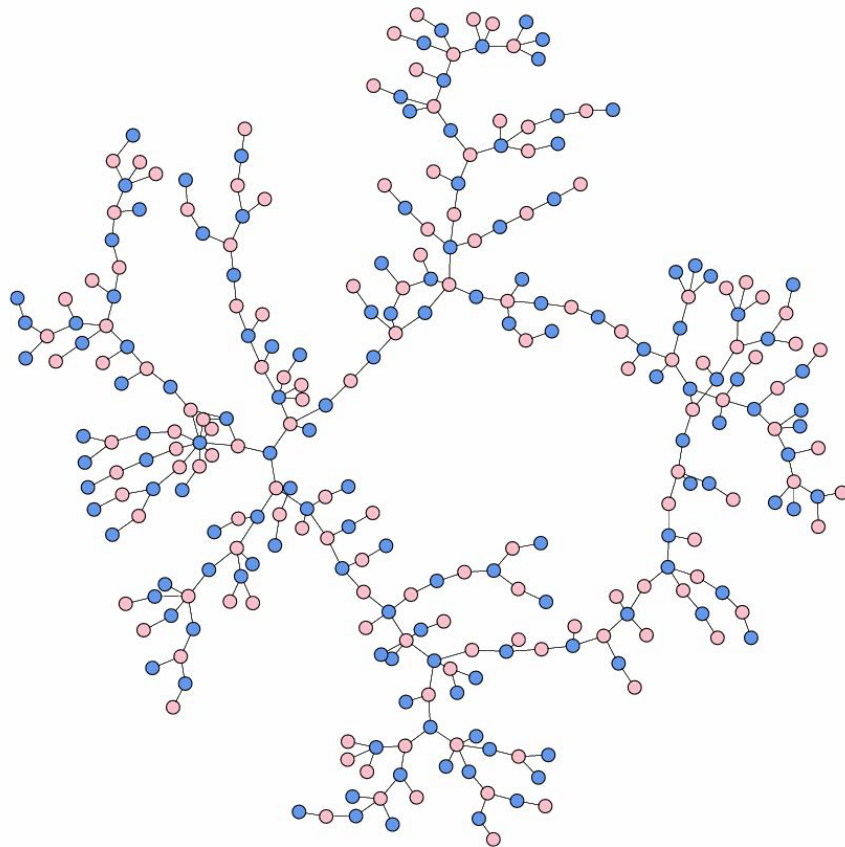
If the fraction of cross-gender edges is significantly more than $2pq$.

Do you remember any example?

Aside: Networks can also exhibit heterophily

If the fraction of cross-gender edges is significantly more than $2pq$.

Yes! The high school dating network

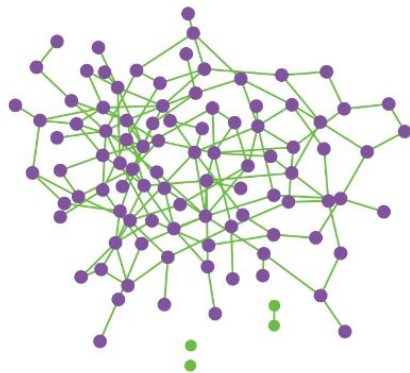
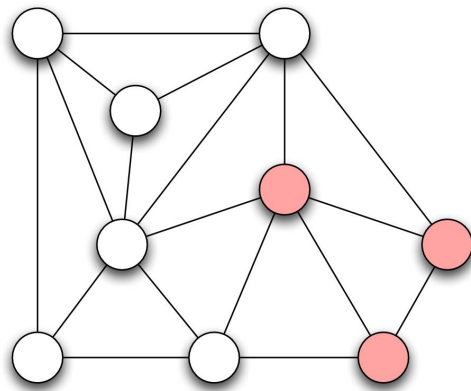


Comparing Homophily between Groups

Problem:

If groups X and Y have different levels of homophily, how can we measure them separately and compare them to each other?

Approach 1: Compare the observed probability of a red-red tie to a random baseline, do the same for the white-white tie, and see which observed probability deviates farther from random.



Measuring homophily

What is the observed probability of a tie between two nodes from group x?

$$\rightarrow \widehat{Pr_{xx}} = \frac{L_{xx}}{L}$$

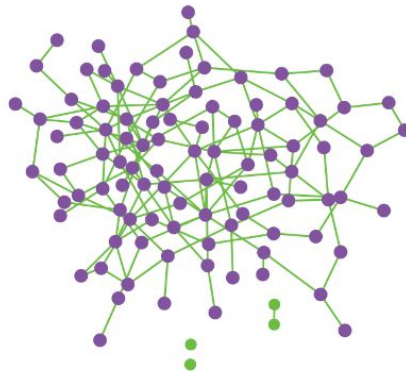
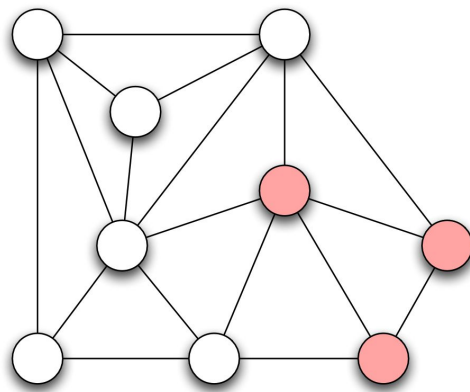
What is the random baseline probability?

→

H₀: $Pr_{xx} = \left(\frac{N_x}{N}\right)^2$ Does the observed deviate from the random baseline?

→

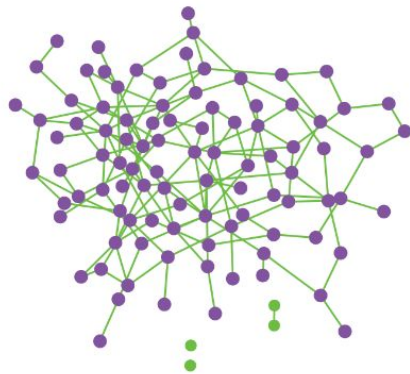
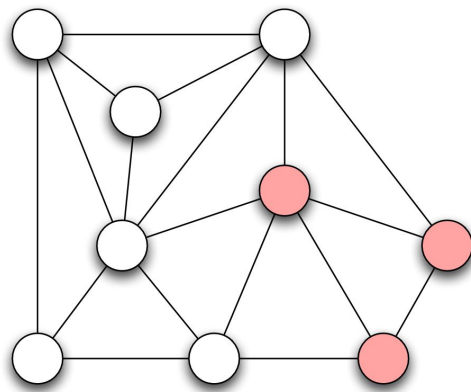
$$\theta_{xx} = \frac{\widehat{Pr_{xx}}}{Pr_{xx}}$$



Measuring homophily

Q: What is a hidden assumption in this homophily test?

Hint: Recall how Erdos-Renyi random graphs are constructed.



Measuring homophily

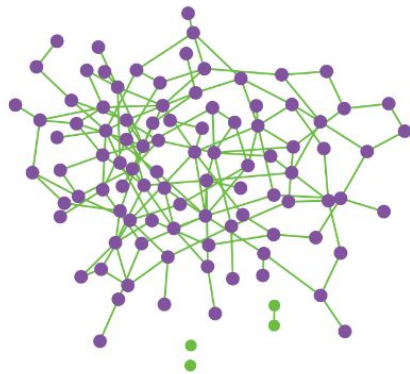
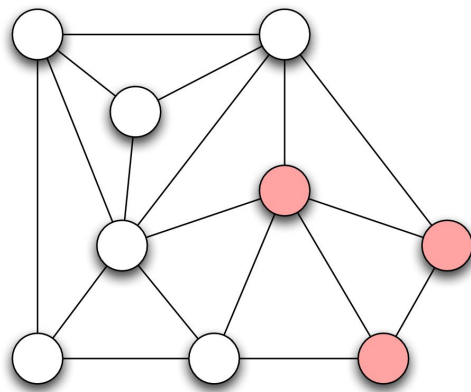
Q: What is a hidden assumption in this homophily test?

Hint: Recall how Erdos-Renyi random graphs are constructed.

Every dyad has equal probability, p , of getting connected

So, **both groups will have the same average degree**

$$D_x = D_y$$



Measuring homophily

Actual average degrees of x and y

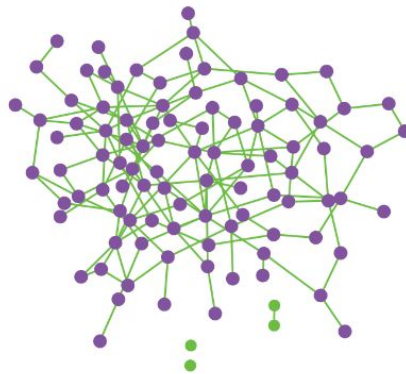
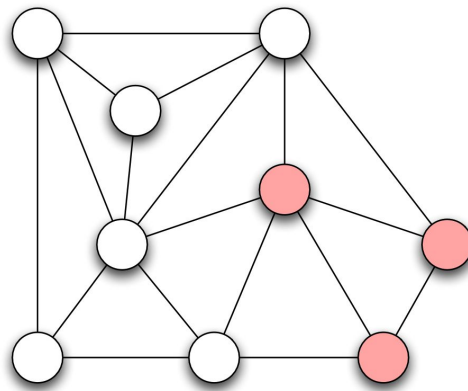
x: red, y: white

$$D_x = 10/3 = 20/6$$

$$D_y = 25/6$$

$$D_x < D_y$$

So, even if group x and y have the same homophilous tendency, group y will have more friends, so they may appear more homophilous



Measuring homophily

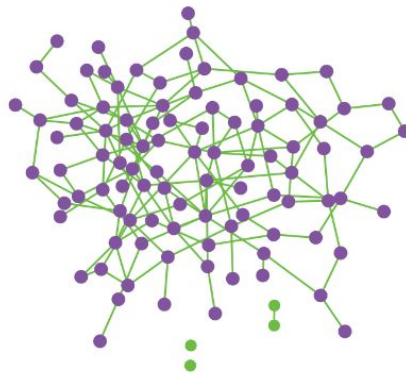
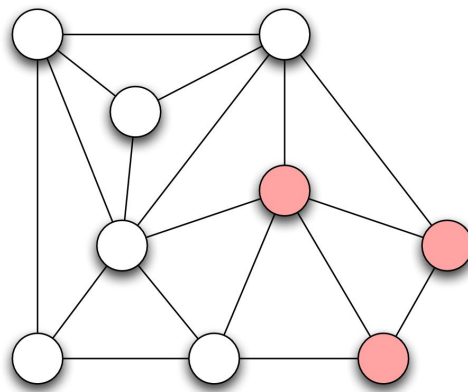
Approach 2

Randomly shuffle the node colors

Approach 3

Randomly shuffle the edges of each node, while holding their degree constant (e.g., $A \rightarrow B, C, D$ to $A \rightarrow E, F, G$)

Q: Which approach solves the problem of different average degrees?



Summary

We've seen another fundamental property of networks: similarity between neighbors

(Recall short paths connecting nodes and triangles formed by common neighbors)

Two extremely powerful analysis techniques: comparison to a random (shuffled) network and longitudinal analysis!